

FERMION CARRIER-GENERATION STRUCTURE I

Premetric Exact-Three Root Structure and the Generation-Governed Standard-Model Matter Bundle

Primitive generation standing, a complete neutral-root receipt,
and a one-Higgs rooted response atlas

The original generation work has exactly three primitive positions. A complete neutral-source receipt identifies those positions with three rooted incidences. The count-free Standard-Model matter profile is instantiated over that base, yielding four typed fibres. Their response slots retain complete source, continuation, and failure data through one Higgs bridge/fixation carrier. Metric and terminal realizations receive this structure through explicit maps.

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Abstract

The primitive generation theorem determines the architecture that jointly performs the original matter, charge, weak, comparison, boundary, and closed-circuit work. This paper receives that theorem into a neutral-root interface and develops its four-kind matter and Higgs-response consequences. The adequacy question is posed before a count: one common construction must perform all original duties, and primitive standing excludes an adequate proper subarchitecture. The resulting architecture has exactly three positions. A complete neutral source interface realizes those positions as distinct rooted incidences; its source-root identification transfers the count to an unlabelled set I_3 with external S_3 action.

Source formation, complete semantic-use signatures, joint occupation, and root-preserving realization are separate steps. The count is neither the number of whole presentations nor the union of roots from mutually incompatible constructions. The integral circuit register has rank one as a consequence of the proved three-position architecture. A larger realized system may carry additional data while having no additional primitive generation standing for the unchanged original work.

The count-free Standard-Model representation-response profile supplies four distinct matter kinds. Base change gives the structural bundle $\mathcal{E}_{\text{SM}} = I_3 \times \{r_N, r_L, r_U, r_D\}$. Each kind fibre is a copy of the same generation architecture; kind signatures and actual physical responses remain distinct. On complete positioned, rooted-realization, and response interfaces satisfying the occupation, witness, and connectivity contracts, the single Higgs bridge/fixation carrier supports the exact typed correspondence developed below. Markings form an S_3 -torsor, cyclic orientations a C_2 -torsor, and starts a C_3 -torsor. Complete-target naturality and explicit structural erasure govern cross-kind comparison. Numerical Yukawa values, pole masses, heavy neutral states, and terminal physical bearers retain their own receiving obligations; none supplies an additional primitive generation root.

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1 Introduction

The generation architecture precedes its metric display. A table of named particles already contains a family count, a representation assignment, and often a mass ordering. To derive the architecture, the work to be performed must instead be stated at open multiplicity. The primitive generation construction does this by fixing the matter and comparison duties while leaving the occurrence set undetermined [3, Definition 2.1 and Sections 3–6]. Its exact result is

$$\text{Prim}_{P_0}(I) \iff |I| = 3.$$

Here primitive standing means that the architecture supports one joint construction of all original duties and no proper subarchitecture does. It concerns which architecture can perform the original work, before additional realization parameters become independently specified answers.

This paper constructs the receiving interface required to use that result in the fermion carrier–generation structure programme. It identifies each primitive position with one neutral source-root incidence, proves complete occupation, and then derives the common three-position matter bundle. Its dependency order is

$$\begin{aligned} \text{original generation work} &\prec \text{primitive standing} \\ &\prec \text{complete neutral-root receipt} \\ &\prec I_3 \prec \text{typed matter fibres} \\ &\prec \text{rooted response slots} \\ &\prec \text{metric and terminal projections.} \end{aligned}$$

No later spectrum or measured family count selects the earlier architecture.

Three distinctions organize the argument. A presentation is not an occurrence: several presentations may express one incidence and a whole presentation may contain several incidences. An occurrence interface must therefore be formed before counting components of a realization groupoid. Second, one joint construction is stronger than separate compatible-looking examples. All original duties, root occupations, domains, and failures must belong to one common packet. Third, additional physical data do not automatically add primitive work. An enlarged realized packet can have irreducible dynamics without acquiring a fourth primitive position for the unchanged original generation task [3, Theorem 6.4].

1.1 Main theorem

Write $\mathbf{F} = \{N, L, U, D\}$ for neutral weak, charged colorless, up-type colored, and down-type colored matter kinds, and $\mathfrak{F} = \{r_N, r_L, r_U, r_D\}$ for their count-free role profile.

Theorem 1.1 (FCGS-I structural correspondence). *The primitive generation construction, complete neutral-root receipt, and count-free role profile give conclusions 1–3 and the position atlas in 5. The realized occurrence and response chain in conclusion 4 uses the complete receiving contracts F1–F9, H1–H7, and R1–R9 specified below:*

1. *the complete neutral-root receipt identifies the original primitive architecture with an unlabelled three-element set I_3 ;*
2. *the generation-governed matter bundle is $\mathcal{E}_{\text{SM}} = I_3 \times \mathfrak{F}$;*
3. *each $p_f : \mathcal{E}_f = I_3 \times \{r_f\} \rightarrow I_3$ is an S_3 -equivariant bijection;*
4. *the source-formed structural and response-slot objects admit the typed chain $\text{Disc}(\mathcal{E}_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y = \text{Disc}(\mathcal{A}_f)$ through the one bridge/fixation carrier $H_{\text{br}/\text{fix}}$; and*

5. the tagged position atlas is $\text{Disc}(\mathbf{F}) \times I_3$, with untagged base-relative codescent I_3 .

The correspondences and their domains are proved in the following sections. No preferred marking, orientation, start, or premetric mass order is selected.

The root theorem is Theorem 4.5. The later fibre, rooted response, and naturality theorems supply the remaining clauses. The structural count is prior to physical enrichment. In particular, neutral propagation and heavy-neutral realization keep their full physical content, but their spectra do not supply a new count of primitive generation work.

2 Fixed target and receiving order

2.1 The count-open original work

Let $W = P_0$ denote the original work of the primitive generation theorem [3, Definition 2.1 and Section 2.1]. It comprises the inherited representation, charge, and weak types (R, Q, W) ; compatible response slots M ; complete direct comparison F ; one coherent source–propagation–readout boundary model B ; a non-removable circuit capability Ω with an admitted witnessing use; count-independent, covariant formation S ; and one common construction of all duties J . The inherited source and carrier licenses are part of the task. A displayed matrix is not a proof of those licenses.

Define $A_{P_0}(I)$ to mean that the architecture I admits one joint construction of this complete work, and define

$$\text{Prim}_{P_0}(I) := A_{P_0}(I) \wedge \forall J \subsetneq I \neg A_{P_0}(J).$$

The quantifier in A_{P_0} permits the existential realization parameters to be supplied together on the architecture being tested. It does not hold all parameters of a previously chosen larger physical packet fixed. This is the original primitive construction problem, not a question about extracting dynamically autonomous pieces of a fixed system.

Complete direct comparison is a substantive part of P_0 . A sparse cycle can have a larger minimal support and therefore requires a different normal-form argument. Similarly, a new prescribed numerical answer, infinitary obligation, or eventual-success requirement belongs to a richer task unless its original status and derivability are proved.

2.2 The neutral receiving target

Fix a source-licensed joint packet of P_0 on I . The neutral receiving target T_N^- retains that packet's original-work identity and the source, boundary, continuation, and failure contracts for its neutral weak multiplicity space. The superscript indicates that empirical family names, mass ordering, detector selections, and terminal-bearer claims have not been used to form the roots. The neutral weak source receives the multiplicity of the same joint generation work. It is not asked to supply a separate neutral-only CP count.

For every nondegenerate determinate target, admissibility, standing, reference, and irreversible fixation are necessary work. Accordingly the kernel governs the packet, its roots, and its receiving incidences. The following source contracts specify which objects perform that work; they do not make kernel governance optional.

The structural receipt and physical realization are distinguished by their maps. Each kind has its own multiplicity space and its own tensor actions. An abstract common generation index supplies no physical pairing of quark and lepton mass eigenlines. Such a pairing would require a further pairing-sensitive source map. Whole-packet transports carry all actual operator data, boundary domains, and failures with them.

2.3 Imported interfaces and mathematical obligations

The cardinality paper proves count-open joint realization at every admitted finite phase class, faithful comparison, and finite generated-use preservation [3, Theorem 4.2 and Propositions 4.3–4.5]. Its Theorems 5.4 and 6.2 prove the exact primitive endpoint. The latter theorem also excludes infinite primitive architectures for this original work. Corollary 5.5 derives rank one after the count. Theorems 6.4 and 6.6 and Proposition 6.7 govern enlargement, actual receipts, and additional reader descent. These results supply the architectural theorem used here.

Role-Occupancy Closure supplies candidate, preassignment, joint compatibility, and complete occupation interfaces [1]. Standing Carrier Identity Closure supplies complete-signature identity and continuation congruence [2]. The matter, charge, and bridge sources supply the count-free types and response licenses [12, 13, 15]. The receiving construction below verifies the root identity and coverage needed to connect those interfaces. No import identifying a complete classifier is treated as a proof of a unique occupant or of a circuit rank.

Three checks remain explicit throughout: every root comes from the declared source inventory; all roots participate in one complete assignment; and each downstream reader preserves its actual domain and answer. A failure of one of these conditions is a concrete failure of the proposed receiving map, even if its displayed set happens to have three elements.

3 Complete neutral occurrence formation

We first specify source identity and semantic uses, then construct the rooted candidate set, then prove joint occupation. A groupoid of whole presentations is never used as a substitute for the occurrence set.

3.1 Complete groupoid-valued use algebra

Let $\tilde{\mathcal{P}}_N$ be the groupoid of source-licensed presentations at the fixed neutral receiving target. Its registered use algebra includes the primitive coordinate functors, continuation-prefix functors, finite products, lawful postcomposition, finite aggregates, typed disposition branches, and consumer projections. Every generator carries its domain and failure outcome. Required successful uses retain their readiness tests. Write (f, v) for a generator and its licensed continuation, and put

$$\mathcal{W}_N = \prod_{(f,v)} W_{f,v}, \quad \Sigma_N^- : \tilde{\mathcal{P}}_N \longrightarrow \mathcal{W}_N.$$

The product signature retains the actual object and arrow values, source incidences, readiness, and failures. Its image is not merely a table of isomorphism classes: the complete groupoid information is retained.

Theorem 3.1 (Complete-use factorization). *Every use generated by the declared algebra factors through Σ_N^- , with the inherited naturality and domain data. Two presentations with the same complete signature have the same interpreted result and disposition under every generated use. Conversely, a difference in a primitive registered coordinate is a separating licensed use.*

Proof. A generator is its own signature projection. Continuation prefixes are registered coordinates. Products and finite aggregates collect the factorizations of their constituents; lawful postcomposition and consumer projection compose them; a typed disposition branch reads the retained domain and failure value. Induction on the finite expression proves the claim, including equality of admission domains. A coordinate projection separates any two signatures differing at that coordinate. Uniqueness

is relative to the actual image of Σ_N^- , not to unused objects of the ambient product. Naturality follows from the registered generator squares and preservation by composition. $\square$

This is complete for the specified generated repertoire. An additional physical continuation must be registered and shown to factor; calling the existing signature complete does not prove this extra fact.

3.2 Canonical activity and congruence

An anchor-complete finite presentation retains the source and target identities and all original-use fields. Among its subpresentations with the same complete signature, order first by the number of nonanchor incidences and then by a fixed canonical presentation code. The code breaks ties among presentations; it selects no physical root or orientation. Work in the source class where at least one finite anchor-complete representative exists. Root identity below remains defined by the source incidence rather than by the chosen display code.

Theorem 3.2 (Deletion-essential active core). *Such a presentation has a signature-equivalent finite core in which no nonanchor incidence can be deleted while preserving the complete signature.*

Proof. The nonempty finite collection of subpresentations has a least incidence count and a least code at that count. Deleting a nonanchor incidence with the same complete signature would give a strictly smaller admissible count, a contradiction. This minimality concerns presentation activity; it is distinct from primitive architecture minimality across joint constructions. $\square$

Theorem 3.3 (Signature congruence before counting). *Complete-signature identity is respected by every registered continuation and generated use. Replacing a whole presentation by an equivalent active core changes no retained rooted incidence or original answer.*

Proof. The factorization theorem supplies continuation and use congruence. Root-incidence coordinates are retained in the signature, so replacing a presentation with that same signature preserves them. In particular, it does not identify two distinct incidences that occur within one presentation. An arbitrary root-erasing isomorphism is not an identity of the retained rooted signature. $\square$

Write $\mathcal{P}_N$ for the resulting active presentation groupoid. This is a validation object. Its component count is not a generation count.

3.3 The source-root receipt at open multiplicity

The joint packet of the cardinality construction has a lepton multiplicity space $V_L = \mathbb{C}^I$ and a positive simple charged response C on it. Let

$$E(C) = \{E : E = E^* = E^2, \text{ rank } E = 1, [E, C] = 0\}.$$

These are the intrinsic boundary projections for its neutral weak source and detection channels. They are distinct from the quark comparison projections of A . At this stage $|I|$ is open. The source license admits the weak components of these channels, the common propagation kernel $U(t)$, and the readout

$$b_{EF}(t) = \text{Tr}(FU(t)EU(t)^*)$$

on its declared parameter domain. No conservation of the boundary label in the interior is asserted.

Definition 3.4 (Complete neutral source-root receipt). *A receipt of a joint P_0 packet consists of its licensed boundary inventory and all rooted neutral weak incidences o_E at $E \in \mathbf{E}(C)$. Equality of roots retains the intrinsic projector incidence, its weak type, and the source target. Whole-presentation duplicates are quotiented using the complete signature. The candidate universe is*

$$\text{Cand}_N = \{o_E : E \in \mathbf{E}(C)\}.$$

An eligible preassignment is an injective, target-covering assignment into this source-formed universe. Its joint-compatibility predicate JC_N requires one common packet, orthogonality and resolution of the boundary identity, all declared original comparison and boundary uses, and the complete readiness/failure record. Define

$$\text{Occ}_N = \bigcup_{\tau: JC_N(\tau)=1} \text{im } \tau.$$

All assignments in this union belong to the same received joint construction and its lawful presentations. Different physical packets are not pooled into one occurrence set.

The source candidate definition contains no numeral and no rank-one circuit premise. Rank-one projectors mean individual source lines at arbitrary multiplicity. A superposed preparation or a derived field remains a use or a realization of this interface; it is not by that fact an additional primitive generation root.

Definition 3.5 (Joint receipt contract). *The receipt contract C_N retains: the original P_0 work and its joint packet; the complete boundary source inventory; the generated-use and failure signature; rooted candidate identity; one complete jointly compatible assignment; and all lawful transports of these data. For the primitive target it additionally retains $\text{Prim}_{P_0}(I)$. The complete comparison graph and phase family belong to the original generation packet. Their transport to the neutral root index is structural index transport, not an identification with every neutral propagation phase.*

Theorem 3.6 (Constructed neutral-root receipt). *For the source-licensed count-open joint packets of the cardinality paper, the construction above has a complete positive assignment and a bijection*

$$\sigma_N : I \xrightarrow{\sim} \text{Occ}_N.$$

It preserves the original generation incidence, the declared boundary domains, and the complete generated-use/failure record. Every finite admitted comparison phase class has such a simultaneous receipt. The primitive original work has a nonempty family of these receipts.

Proof. In the displayed packet $C = \text{diag}(c_i)$ has distinct positive c_i on $\mathbb{C}^I$. If a rank-one projector E commutes with C , its one-dimensional image is a C -invariant line and hence an eigenline. Conversely each eigenline projector commutes with C . Thus $\mathbf{E}(C) = \{E_i : i \in I\}$, the projectors are pairwise orthogonal, and $\sum_i E_i = 1$. The source-root map is $i \mapsto o_{E_i}$. Injectivity follows from retained projector incidence; every candidate is one of these roots, proving source exhaustiveness before a numerical count.

The full assignment covers all candidates together. In the common neutral packet $N = \text{diag}(1, \dots, n) + \mathbf{11}^T$, $U(t) = e^{-itN}$, the bound

$$|U_{ji}(t)| \geq t - 2n^2t^2 \geq t/2 > 0 \quad (i \neq j, 0 < t \leq 1/(4n^2))$$

supplies simultaneous boundary participation at open finite multiplicity [3, Theorem 4.2]. The same theorem supplies the charged, gauge, charge, weak, and bridge witnesses and every declared comparison phase class in the same packet. There is no use of independently chosen channel witnesses. Propositions 4.3 and 4.5 of that source preserve the original phase quotient, typed generated uses, actual

domains, and failures under lawful transport. Consequently the full assignment satisfies JC_N . Its image is all candidates, while every other positive preassignment has image inside that set. Hence the live union is exactly Cand_N .

Simple-spectrum covariance sends E_i to VE_iV^* with every other neutral tensor and source map transformed according to its type. Relabeling I transports the assignment. There is no preferred physical marking and no induced pairing-sensitive quark–lepton map. Finally, the primitive construction and explicit common witness of the cardinality paper supply at least one primitive packet, proving nonvacuity. $\square$

The theorem constructs the receipt on its stated source inventory. A new physical source inventory must establish the same complete occupation and use preservation before it invokes this root identification. Its mere possession of three display coordinates does not suffice.

3.4 The vertical realization groupoid

Definition 3.7 (Neutral vertical realization groupoid). *The objects of $\mathcal{R}_N^{\text{vert}}(C_N)$ are triples (o, a, j) , with $o \in \text{Occ}_N$, $a \in \text{Ob}(\mathcal{P}_N)$, and a certified rooted realization $j : o \hookrightarrow a$. The certificate includes equality with the fixed received packet’s complete rooted signature and a licensed root-preserving identification with that signature’s canonical source core. A realization with a different response, boundary, or continuation signature is a different received packet, not another object of this fixed fibre. Its arrows are complete-signature, continuation-natural isomorphisms over the identity of the retained root. Its structural zero-truncation is $\mathcal{S}_N(C_N) := \tau_{\leq 0}\mathcal{R}_N^{\text{vert}}(C_N)$: it retains the connected components as a discrete groupoid.*

Theorem 3.8 (Occurrence–realization comparison). *The projection $q_N : \mathcal{R}_N^{\text{vert}}(C_N) \rightarrow \text{Disc}(\text{Occ}_N)$ has nonempty connected fibres and induces $\pi_0\mathcal{R}_N^{\text{vert}}(C_N) \cong \text{Occ}_N$. Consequently $\mathcal{S}_N(C_N) \simeq \text{Disc}(\text{Occ}_N)$.*

Proof. The complete positive assignment supplies a realization of every root. For each certified presentation, the retained identification gives an actual licensed map to the fixed rooted source core. Composing one such map with the inverse of the other gives a root-preserving zigzag. The source identity contract [2] supplies these maps on its complete-signature domain; equality of the root name alone is not used to infer them. Distinct roots have no connecting arrow, because arrows lie over the identity. Thus components are exactly the already formed roots. The component bijection becomes the stated structural equivalence after zero-truncation. No contractibility of the untruncated source fibres is inferred from connectedness. $\square$

4 The exact-three root theorem

The count theorem applies to the original primitive work and then to its complete neutral receipt. It does not infer primitive minimality from the absence of a displayed fourth particle.

Lemma 4.1 (Nonzero capacity requires three positions). *If $A_{P_0}(I)$, then I has at least three members.*

Proof. A complete simple graph on fewer than three vertices has no closed non-removable comparison circuit. It therefore cannot perform the Ω duty of P_0 [3, Theorem 3.2]. $\square$

Lemma 4.2 (Complete-comparison transport). *Transport of the original direct-comparison graph through σ_N gives the complete simple graph Γ_N on Occ_N .*

Proof. Every pair of distinct original positions has its declared direct comparison by the F duty. The bijection maps distinct pairs exactly to distinct pairs of roots. Thus it preserves and exhausts all edges. No appeal to the absence of a selector is needed to infer an edge. $\square$

Lemma 4.3 (Integral circuit quotient). *For a finite connected graph Γ , $C^1(\Gamma; \mathbb{Z})/dC^0(\Gamma; \mathbb{Z})$ is free of rank $|E| - |V| + 1$. In particular its rank for K_n is $(n - 1)(n - 2)/2$.*

Proof. Choose a spanning tree. Integer vertex coboundaries set its $|V| - 1$ coordinates to zero, uniquely modulo a constant vertex value. Each remaining edge then contributes one freely variable integral coordinate. $\square$

Lemma 4.4 (Faithful original circuit family). *The original generation phase family, indexed on Occ_N through σ_N , retains every original loop consequence and its reversal law.*

Proof. The upstream intrinsic readers $\text{Tr}(P_i B P_j B P_k B)$ separate the phase quotient and transform covariantly with the complete incidence [3, Proposition 4.3]. Reindexing by a bijection changes neither their domains nor their values. This is an assertion about the transported original generation comparison. The admitted neutral propagation witness may be real; its own CP-sensitive outcomes are retained independently. $\square$

Theorem 4.5 (Exact three from primitive standing). *Every complete neutral receipt of the original primitive generation work satisfies*

$$|I| = |\text{Occ}_N| = |\pi_0 \mathcal{S}_N(C_N)| = 3.$$

The result selects no marking or embedded subpacket of a fixed larger physical realization.

Proof. The cardinality theorem proves $\text{Prim}_{P_0}(I) \iff |I| = 3$ [3, Theorem 6.2]. Its upper bound uses complete comparison: a non-removable circuit has a live triangle, and the uniform joint receiver realizes every original duty on that support. An architecture with more than three members, including an infinite one, therefore has an adequate proper subarchitecture and has no primitive standing for this work. The lower bound is Lemma 4.1. Theorem 3.6 gives the first receiving bijection; Theorem 3.8 gives the component bijection. Composing them proves the claimed counts. The constructional proper-subarchitecture argument supplies its existential parameters jointly; it asserts no compression or autonomous restriction of an arbitrarily fixed dynamics. $\square$

The integral circuit register consequently has rank $(3 - 1)(3 - 2)/2 = 1$. We write Ω_N^{prim} for this transported primitive generation register. Its rank follows from the architecture, and is not an additional counting axiom.

Definition 4.6 (Derived generation base). *Set $I_3 := \text{Occ}_N$. Its complete incidence graph is K_3 and $\text{Aut}(K_3) \cong S_3$ acts externally on it, together with transport of all role-indexed queries.*

The quotient I_3/S_3 is a singleton, but it is not the generation base. The external action expresses absence of preferred names while retaining three distinct rooted incidences. It is not a physical symmetry exchanging unequal measured masses.

Corollary 4.7 (No additional primitive generation root). *An enlarged architecture with more than three occupied positions has no primitive standing for unchanged P_0 . Positions below three cannot perform the original work. Additional realization data do not enlarge the primitive root set under an original-work-preserving receipt.*

Proof. The first two claims are the exact equivalence used in Theorem 4.5; the last is its transport through a complete receipt [3, Theorems 6.4 and 6.6]. Additional readers descend only when constant on the actual nonempty receipt fibres and equal to the specified receiving answer [3, Proposition 6.7]. $\square$

Dependency and scope. Original work and primitive standing determine the count. Source formation and complete joint occupation transfer it to neutral roots. The one-dimensional integral circuit register follows after the count. Metric responses, heavy states, and terminal objects enter through their own typed receiving maps.

5 The count-free Standard-Model matter profile

Theorem 4.5 supplies a generation base. It does not by itself say which matter roles are instantiated over that base. The next input is the generation-erased projection of the Standard-Model representation-response structure established in [15]. Only its typing is imported, not the family count recorded in its envelope.

5.1 Generation erasure and four composite matter roles

Let Σ_{SM} denote the structural tuple. Define

$$\pi_{\text{cf}} : \Sigma_{\text{SM}} \longrightarrow \Sigma_{\text{SM}}^{\text{cf}}$$

by deleting N_g , every family name and index, every family-indexed matrix dimension, and every family-indexed numerical coordinate. It retains gauge-representation, color, weak-component, relative-charge, chirality, bridge-identity, operator-status, boundary, and continuation types. Only $\Sigma_{\text{SM}}^{\text{cf}}$ is used below, after Theorem 4.5.

The chiral structure contains a left quark doublet, up- and down-type singlet response types, a left lepton doublet, and a charged-lepton singlet response type. Its composite matter-response classes are

$$\begin{aligned} r_N &= \rho_{\text{mat}}(L_L^0, O_\nu), & r_L &= \rho_{\text{mat}}(L_L^-, e_R, O_e), \\ r_U &= \rho_{\text{mat}}(Q_L^u, u_R, O_u), & r_D &= \rho_{\text{mat}}(Q_L^d, d_R, O_d). \end{aligned}$$

The map ρ_{mat} retains those complete coordinates and forgets only field-presentation skin. Write $\mathfrak{F} = \mathfrak{F}_{\text{cf}} = \{r_N, r_L, r_U, r_D\}$ and $\mathbf{F} = \{N, L, U, D\}$. These are the neutral and charged members of the lepton weak pair and the up and down members of the colored weak pair: structural role classes, not particle names or generations.

The table summarizes the role coordinates certified by the generation-erased Standard-Model structural tuple [15, Def. 1.7 and the field-role table following it]. Color, weak incidence, and charge are witness-bearing coordinates. The displayed electric charges are not inserted as separate primitives; they are the conventional normalization of a relative hypercharge lattice.

5.2 Charge and Higgs-operator separation

For one chiral profile, Yukawa compatibility and anomaly cancellation give

$$Y_L = -3Y_Q, \quad Y_e = -6Y_Q, \quad Y_H = 3Y_Q, \quad Y_u = 4Y_Q, \quad Y_d = -2Y_Q,$$

up to an overall nonzero normalization and simultaneous sign reversal [13, Lemmas 9.1–9.2 and Thm. 9.3]. The relations are compatible with the operator-status types

$$\overline{Q}_L \tilde{H} u_R, \quad \overline{Q}_L H d_R, \quad \overline{L}_L H e_R.$$

Table 1: Count-free matter-role profile. The listed charges are a conventional normalization of the rigid relative charge class.

Role	Color	Weak incidence	Charge	Structural Higgs-response coordinate
r_N	singlet	neutral lepton-pair member	0	neutral weak response-status coordinate; operator may be absent
r_L	singlet	charged lepton-pair member	-1	charged-lepton H operator-status coordinate
r_U	triplet type	up member of quark pair	$2/3$	up-type $\tilde{H}$ operator-status coordinate
r_D	triplet type	down member of quark pair	$-1/3$	down-type H operator-status coordinate

Proposition 5.1 (Pairwise role separation). *No admissible redescription preserving the complete matter target identifies two distinct elements of $\mathfrak{F}$.*

Proof. The neutral and charged colorless roles differ in charge-response and weak component. A colorless role and a colored role differ in color-response type. The two colored roles differ in weak component, relative hypercharge, and $H/\tilde{H}$ incidence. Overall normalization, global sign, gauge basis, and notation change none of those relative mismatches. An identification would therefore erase a standing-bearing coordinate or move to a different target, neither of which is an admissible same-target redescription. $\square$

The matter-carrier coverage theorem excludes an additional same-scope primitive matter kind outside the certified representation, color-bearing, charged-colorless, neutral-weak, and mass-response roles [12, Thms. 19.3–19.11]. The mass-response slot is an incidence carried by matter rather than a fifth fermion kind. At the later global carrier level, no residual same-scope primitive survives outside the certified package [16, Thms. 17.1 and 20.1]. These are role-type exhaustion results; they do not supply a second generation count.

6 Generation-governed base change

Generation cardinality has a precise source meaning: it is the cardinality of the repeated copies of an already typed representation-family profile [14, Def. 9.1]. Changing that cardinality changes field-role instances, response and parameter slots, and the mixing and boundary ledgers; it is not a presentation coordinate [14, Prop. 9.2]. Only that typing rule is imported, not the source envelope’s empirical value. The primitive-standing theorem and complete neutral occupation receipt supply the structural value and its actual unlabelled index set [3, Thms. 5.4 and 6.2].

Definition 6.1 (Generation extension). *For any finite set I , define the generation extension of the matter profile by*

$$\text{Gen}_I(\mathfrak{F}) := I \times \mathfrak{F}.$$

The first coordinate is the generation root and the second is the matter-role type. No quotient identifies different first or second coordinates.

Definition 6.2 (FCGS matter bundle). *Using the independently derived set I_3 , define*

$$\mathcal{E}_{\text{SM}} := \text{Gen}_{I_3}(\mathfrak{F}) = I_3 \times \mathfrak{F}.$$

For $f \in \mathbb{F}$, let

$$\mathcal{E}_f := I_3 \times \{r_f\}$$

and write

$$p_f : \mathcal{E}_f \longrightarrow I_3, \quad p_f(i, r_f) = i.$$

This construction occurs after the count. It instantiates the role profile over the derived base rather than manufacturing new source IDs and then using their number to justify the base.

Theorem 6.3 (Post-count four-kind representation theorem). *For every $f \in \mathbb{F}$, the projection p_f is a canonical S_3 -equivariant bijection. Consequently*

$$\mathcal{E}_N \cong \mathcal{E}_L \cong \mathcal{E}_U \cong \mathcal{E}_D \cong I_3$$

as unlabelled generation bundles, and

$$|\mathcal{E}_f| = 3, \quad |\mathcal{E}_{\text{SM}}| = 12.$$

Proof. The map

$$s_f : I_3 \longrightarrow \mathcal{E}_f, \quad s_f(i) = (i, r_f)$$

is inverse to p_f . Thus p_f is a bijection without a chosen numbering. For $\sigma \in S_3$, define $\sigma \cdot (i, r_f) = (\sigma i, r_f)$. Then $p_f(\sigma \cdot (i, r_f)) = \sigma p_f(i, r_f)$, establishing equivariance. The four fibres are disjoint because the role coordinate is retained and Proposition 5.1 forbids a target-preserving identification of distinct roles. The cardinalities follow from $|I_3| = 3$. $\square$

The number twelve counts generation–role positions in this structural bundle. It is not a count of color states, spin components, antiparticles, confined hadrons, or terminal physical particles. Those belong to different targets.

Corollary 6.4 (Quark-generation structure). *The up-type and down-type quark fibres have the same derived three-root generation structure:*

$$\mathcal{E}_U \xrightarrow[p_U]{p_U} I_3 \xleftarrow[p_D]{p_D} \mathcal{E}_D.$$

They remain inequivalent typed fibres.

Proof. The common structure is Theorem 6.3. Inequivalence follows from Proposition 5.1: the U and D signatures differ in relative charge, weak member, and conjugate-Higgs incidence even though their root coordinates range over the same I_3 . $\square$

The shared graph K_3 and phase register transport to each kind as $p_f^* K_3$ and $p_f^* \Omega^{\text{prim}}$. They describe the common generation architecture, not independent native rank-one holonomy in every species. The neutral-root receipt transports the joint generation register; it does not require a nonzero measured neutral CP phase.

6.1 Post-count positioned occurrence formation

For $i \in I_3$, let $\mathfrak{F}_i = \{i\} \times \mathfrak{F}$ be the positioned one-cell profile. Live occupants satisfy the complete nine-coordinate eligibility vector, joint compatibility, deletion essentiality, weak-pair involution, bridge-slot compatibility, and charge/representation compatibility. Let R_f be the fixed kind- f realization target

obtained by positioning this cell over I_3 , and C_f its complete activated interface: target, semantic uses, complete signature, joint compatibility, continuation, boundary, and total failure data. Define

$$\text{Occ}_f(R_f) = \left(\coprod_{i \in I_3} \text{Live}_f(\mathfrak{F}_i) \right) / \sim_{\text{dup}}, \quad \pi_f : \text{Occ}_f(R_f) \longrightarrow I_3.$$

Duplicate-presentation reduction retains the root (i, f) , which π_f sends to i .

The family-cell transfer uses the complete positioned interface specified by F1–F9 in [5], with the hypothesis map in Appendix A. In particular, the receiver must supply exactly one jointly compatible live occupant over every (i, f) after duplicate reduction (F4), and every licensed kind occurrence must be covered by those cells (F5). The weak-pair, anchor, root-separation, and simultaneous source-cell compatibility conditions are retained. The cited transfer theorem proves the bijection from these contracts; it does not independently establish F4 or F5 for an arbitrary source receiver.

Theorem 6.5 (Exact post-count family-cell transfer). *Under the positioned-interface contracts F1–F9, for every $f \in \mathbf{F}$, π_f is an S_3 -equivariant bijection. The unique occupant over every positioned cell gives a canonical equivariant bijection*

$$u_f : \mathcal{E}_f \xrightarrow{\cong} \text{Occ}_f(R_f), \quad \pi_f u_f = p_f.$$

This transfers the proved base without recounting it or imposing an independent species-specific primitive rank-one mechanism.

Proof. F4 supplies one live occupant over every (i, f) ; π_f is surjective with singleton duplicate-reduced root fibres. F5 supplies complete coverage; this excludes occurrences outside those cells, giving injectivity. Root transport gives $\pi_f(\sigma x) = \sigma \pi_f(x)$. Weak-pair coherence, neutral-anchor agreement, and source-cell compatibility preserve complete incidence data. The map u_f sends (i, r_f) to that unique occupant. These are the conditional positioned family-cell conclusions [5]. $\square$

6.2 Rooted realization and exact generation correspondence

Let $\mathcal{R}_f(R_f)$ be the rooted-realization groupoid. Its objects are triples (a, o, j) of active complete-signature presentation a , occurrence $o \in \text{Occ}_f(R_f)$, and certified rooted incidence $j : o \hookrightarrow R_f(a)$. Arrows are continuation-natural complete-signature isomorphisms transporting j and commuting with every activated consumer. No internal arrow changes root. Put $\mathcal{S}_f(C_f) = \tau_{\leq 0} \mathcal{R}_f(R_f)$.

Assume the rooted-comparison contracts H1–H7 of [6], listed in Appendix A. They require prior activity reduction and congruence, fixed-target restriction, preserved eligibility and compatibility, and external symmetry. H4 supplies an actual rooted-realization witness for every occurrence and registers every such witness. H5 requires two realizations of one registered occurrence to be connected by a licensed root-preserving continuation zigzag. Root separation forbids internal cross-root arrows. H4 and H5 are receiving obligations; the component theorem below derives their zero-truncated consequence.

Theorem 6.6 (Rooted-realization disintegration). *Under F1–F9 and H1–H7, the root projection*

$$q_f : \mathcal{R}_f(R_f) \longrightarrow \text{Disc}(\text{Occ}_f(R_f))$$

has nonempty connected fibres and induces canonical equivalences

$$\mathcal{S}_f(C_f) \simeq \text{Disc}(\text{Occ}_f(R_f)) \simeq \text{Disc}(I_3).$$

The active presentation groupoid itself is not counted.

Proof. H4 provides a rooted realization over every occurrence. H5 connects any two realizations of one root; root separation excludes internal arrows between different roots. Components are thus exactly registered occurrences. The second equivalence is Theorem 6.5; the first is the rooted comparison theorem [6, Secs. 2–4], independent of the active presentation quotient’s component count. $\square$

Let $z_f : \text{Disc}(\text{Occ}_f(R_f)) \xrightarrow{\sim} \mathcal{S}_f(C_f)$ be the inverse canonical zero-truncation equivalence. Define the positioned-fibre map

$$\eta_f : \text{Disc}(\mathcal{E}_f) \longrightarrow \mathcal{S}_f(C_f), \quad \eta_f = z_f \circ \text{Disc}(u_f).$$

With $s_f(i) = (i, r_f)$, its root-normalized form is

$$\eta_f^{I_3} := \eta_f \circ \text{Disc}(s_f) : \text{Disc}(I_3) \longrightarrow \mathcal{S}_f(C_f).$$

Conversely $\eta_f = \eta_f^{I_3} \circ \text{Disc}(p_f)$. Thus the canonical source is the positioned fibre, while displays starting from the common root base use $\eta_f^{I_3}$.

Theorem 6.7 (Exact generation–occurrence correspondence). *Under the contracts of Theorems 6.5 and 6.6, for every $f \in \mathbf{F}$, both η_f and $\eta_f^{I_3}$ are fully faithful, essentially surjective, and externally S_3 -equivariant. Both preserve positioned role, complete eligibility data, rooted incidence, weak-pair edge, charge-compatibility cell, bridge slot, and deletion status. It is the post-count incidence transfer, not a freely chosen isomorphism between two triangles.*

Proof. The factors s_f, u_f, z_f are canonical equivariant equivalences by Theorems 6.3, 6.5, and 6.6. The composites defining both displayed maps have the stated properties. Completeness, pairing coherence, neutral-anchor agreement, root separation, and source-cell compatibility give incidence preservation. $\square$

Physical family alphabets require a marking, canonical only up to the external S_3 torsor. These theorems form positioned L, U, D occurrences after the base is proved, using source-certified occupancy and coverage; they do not independently reconstruct count-blind species registries.

One count, four proved occurrence fibres. Joint primitive standing supplies the architecture; complete neutral occupation identifies its root base. The count-free matter cell is positioned over it. Complete occupancy and rooted-realization disintegration then prove the four occurrence fibres without an empirical family-count premise.

7 The fermionic differentiation module

Retain the positioned and rooted-realization contracts F1–F9 and H1–H7 throughout this section. The derived base supplies three rooted positions in every matter-kind fibre. The complete response arena is formed independently at each root, reduced by rooted response equivalence, and only then compared with the matter fibre. The result is one bridge/fixation carrier supporting three rooted structural Higgs-response-slot incidences per kind. It is not a claim of three Higgs carriers, positive Yukawa couplings, or distinct numerical masses.

7.1 One bridge and the complete rooted response arena

The structural tuple [15] contains the unique bridge/fixation carrier $H_{\text{br/fix}}$ and typed Yukawa-realizability ledger O_Y . The Higgs bridge theorem gives the response-bearing role [18, Defs. 4.1–4.3, 7.1, 8.1 and Thms. 7.4, 7.6, 8.4, 9.2, 12.1]; Higgs identity excludes an additional same-scope carrier [17, Thms. 6.6, 7.3, 8.1, 14.4 and Cor. 8.2].

Fix $f \in \mathbf{F}$ and $o \in \text{Occ}_f(R_f)$. Theorems 6.5 and 6.6 identify o with a positioned role instance without a second count. The receiving response interface carries the rootwise slot-existence contract R4: every such o has an admitted source-mandated structural mass-response object through $H_{\text{br}/\text{fix}}$. This contract requires an actual structural response witness, not merely the root written in a record. It asserts neither a positive coefficient nor a choice among Dirac, Majorana, Weinberg, sterile, obstructed, or operator-absent statuses [7, 10].

Definition 7.1 (Raw rooted response arena). *Let $\mathbf{Y}_f(o)$ be the groupoid of all complete, source-authorized response objects rooted at o , formed before any standing class or quotient representative is selected. Every object records*

*($R_f, o, H_{\text{br}/\text{fix}}, O_Y(f)$, response topology, operator-status domain,
complete witness signature, boundary, continuation, projection flags, failure provenance).*

Its activated branches are ordered and typed. Exactly the standing-obstruction, operator-absent, witnessed-coupling, and exact-rooted-slot branches enter the structural slot quotient. Every admitted object is a complete total response object. An interface-incomplete proposal is retained only as a failure record, never licensed into the standing quotient.

Presentation, basis, rephasing, normalization, texture, chart, regulator, scheme, fit, and duplicate-probe skins define no new responses. Their removal follows completion of the source-authorized witness, continuation, consumer, status, obstruction, and failure registries. This preselection order is required by canonical Yukawa reduction [19, Defs. 3.3–3.4, 5.1, 5.3, 6.1–6.2 and Thms. 5.4–5.5, 6.3–6.5, 7.1, 26.1].

Write $\text{Core}_Y(-)$ for the canonical active response core: first complete the interface, then remove only response-inert presentation skin. The receiving interface also carries R6: complete rooted response equivalence is a continuation congruence and, for every fixed root o , the entire licensed standing fibre $H_f(o)$ is connected. Any two of its admitted objects must be joined by a licensed root-preserving continuation zigzag preserving every registered consumer, status, boundary, and failure answer. The independent response-slot theorem uses this connectivity contract; it does not derive it from bridge identity, a unique classifier, or the desired component count [7, 10]. Different preserved consumer or status answers cannot be silently identified; if they prevent such a zigzag, that receiver has not satisfied R6.

Definition 7.2 (Rooted response equivalence and standing fibre). *Two activated objects of $\mathbf{Y}_f(o)$ are equivalent, $y \sim_f^Y y'$, when their complete response signatures are related by a target-relative continuation isomorphism commuting with every registered response consumer. The relation is defined only within one fixed root. Put*

$$H_f(o) := \text{Core}_Y(\mathbf{Y}_f(o)^{\text{act}}) / \sim_f^Y.$$

This standing root fibre retains the rooted structural slot and complete typed status, including standing-zero, obstruction, and operator-absent branches.

The response interface satisfies R1–R9 of [10]: prior root formation; unique bridge identity; complete preselection registries; rootwise structural witness existence (R4); canonical factorization; rooted congruence and connectivity (R6); root separation; the total ordered branch classifier; and the projection firewall. R4 and R6 must be supplied at this complete fixed rooted interface. They are not established by listing source licenses or invoking unique carrier/classifier identity. Under these contracts the following theorem derives the global component correspondence.

7.2 The independent rooted response-slot quotient

Definition 7.3 (Global rooted slot quotient). *Define the structural coproduct and root projection*

$$Q_f^{\text{slot}} := \coprod_{o \in \text{Occ}_f(R_f)} H_f(o), \quad r_f^Y : Q_f^{\text{slot}} \longrightarrow \text{Disc}(\text{Occ}_f(R_f)).$$

No internal morphism changes the root.

Theorem 7.4 (Independent rooted response-slot quotient). *Under R1–R9, for every $f \in \mathbf{F}$, r_f^Y has nonempty connected fibres and induces a canonical bijection*

$$\pi_0 Q_f^{\text{slot}} \cong \text{Occ}_f(R_f) \cong I_3.$$

Consequently each kind has exactly three rooted structural Higgs-response-slot incidences through the single carrier $H_{\text{br/fix}}$:

$$|\pi_0 Q_f^{\text{slot}}| = 3, \quad \left| \pi_0 \left(\coprod_{f \in \mathbf{F}} \{f\} \times Q_f^{\text{slot}} \right) \right| = 12,$$

while Higgs carrier identity remains one.

Proof. Fix $o \in \text{Occ}_f(R_f)$. The raw arena is formed before selection and is complete at the frozen interface by R3. R4 supplies an activated structural response object, so $H_f(o)$ is nonempty. R5 supplies canonical factorization, and R6 connects any two admitted objects of the entire standing root fibre $H_f(o)$ by a licensed root-preserving zigzag. Connectivity of each signature subclass alone would not suffice. Unique carrier or classifier identity is not a substitute for R6. R7 root separation forbids internal arrows between different summands. Components of the coproduct are therefore exactly registered occurrences. The family-cell theorem identifies those with I_3 ; Theorem 4.5 gives its cardinality. Unique carrier identity fixes every incidence at $H_{\text{br/fix}}$ without identifying its roots. This uses the complete rooted-response theorem [5, 6, 7, 10]. $\square$

The source zero-truncated response quotient is canonically

$$Q_{f,\text{acc}}^Y \cong \text{Disc}(\pi_0 Q_f^{\text{slot}}).$$

Only after Theorem 7.4 define $\mathcal{A}_f := \pi_0 Q_f^{\text{slot}}$. Let $*_o$ be the unique element of $\pi_0 H_f(o)$ and $j_o : H_f(o) \hookrightarrow Q_f^{\text{slot}}$ the coproduct inclusion. With the canonical equivalence $\bar{q}_f : \mathcal{S}_f(C_f) \xrightarrow{\sim} \text{Disc}(\text{Occ}_f(R_f))$, define

$$\chi_f : \mathcal{S}_f(C_f) \longrightarrow \text{Disc}(\mathcal{A}_f), \quad \chi_f(x) := (j_{\bar{q}_f(x)})_* (*_{\bar{q}_f(x)}).$$

Theorem 7.5 (Exact rooted response correspondence). *Under the contracts of Theorem 7.4, for every $f \in \mathbf{F}$, χ_f is fully faithful, essentially surjective, natural on the fixed-target continuation domain, and equivariant for the transported external S_3 action. In particular, $|\mathcal{A}_f| = |\text{Occ}_f(R_f)| = 3$.*

Proof. Theorem 7.4 supplies a canonical bijection on objects. The equivalence $\bar{q}_f$ and discretization make χ_f fully faithful and essentially surjective. A lawful continuation preserves the root, $H_{\text{br/fix}}$, complete response signature, boundary, and continuation; it carries the unique standing class over o to that over the transported root. Define $\sigma \cdot \chi_f(x) = \chi_f(\sigma \cdot x)$; root separation and bijectivity make this well defined. Thus χ_f is natural at the fixed full-target fibre and externally equivariant. $\square$

The raw arenas and standing fibres precede χ_f : it records the quotient theorem rather than manufacturing objects, existence, connectivity, or uniqueness.

7.3 Positive witnesses are a separate subquotient

Let

$$\text{Occ}_f^+ := \{o \in \text{Occ}_f(R_f) : \text{Branch}_f(o) = \text{WitnessedCoupling}\}, \quad Q_f^+ := \coprod_{o \in \text{Occ}_f^+} H_f^+(o).$$

Here WitnessedCoupling is the unique connected positive terminal branch and $H_f^+(o)$ its positive standing fibre. Only that branch enters the positive quotient. A positive but nonunique proposal remains outside it. Standing zeros, compatibility obstructions, operator absences, and exact structural-slot statuses remain in Q_f^{slot} without becoming positive witnesses.

Corollary 7.6 (No positivity promotion). *The structural theorem gives*

$$|\pi_0 Q_f^{\text{slot}}| = 3, \quad |\pi_0 Q_f^+| = |\text{Occ}_f^+| \leq 3.$$

Equality in the second formula requires $\text{Occ}_f^+ = \text{Occ}_f(R_f)$ separately. The first implies neither three nonzero Yukawa couplings nor distinct masses.

Proof. The slot quotient has one connected rooted standing fibre per registered occurrence. The positive quotient is the coproduct over the indicated subset. Its component count is that subset's size. Numerical positivity and inequality are downstream projections absent from the structural slot-existence theorem. $\square$

7.4 Rooted and root-erased degeneracy

Write $Y_f(x) = \chi_f(x)$. For distinct $x, x' \in \text{Ob}(\mathcal{S}_f(C_f))$, define

$$\begin{aligned} \Delta_f^{\text{rooted}} &= \{(x, x') : Y_f(x) = Y_f(x')\}, \\ \Delta_f^{\text{erase}} &= \{(x, x') : \text{EraseRoot } Y_f(x) = \text{EraseRoot } Y_f(x')\}, \\ \Delta_f^{\text{num}}(\rho) &= \{(x, x') : \rho(Y_f(x)) = \rho(Y_f(x'))\}, \end{aligned}$$

where ρ is a separately licensed numerical projection.

Theorem 7.7 (Degeneracy separation). *The rooted structural locus is empty: $\Delta_f^{\text{rooted}} = \emptyset$. The root-erased coupling-signature and numerical loci may be nonempty without merging matter roots or response-slot components or changing exact three.*

Proof. Theorem 7.5 is fully faithful and preserves root, so distinct roots cannot have the same rooted class. Root erasure forgets that coordinate; a numerical projection can map distinct structural responses to the same value. Neither equality lifts to equality in the rooted slot quotient. Conversely numerical splitting creates no element of I_3 . $\square$

Fermionic differentiation is exact at the rooted structural response level: three roots remain three incidences through one carrier. Pairwise distinct root-erased topologies, positive couplings, masses, or mass order are separate.

7.5 When the conventional designations become available

Let $\mathfrak{M} = \text{Bij}(\{1, 2, 3\}, I_3)$. For a marking $m \in \mathfrak{M}$, the composite

$$\text{Disc}\{1, 2, 3\} \xrightarrow{\text{Disc}(m)} \text{Disc}(I_3) \xrightarrow{\eta_f^{I_3}} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} \text{Disc}(\mathcal{A}_f)$$

licenses G_1, G_2, G_3 .

Proposition 7.8 (Designation torsor). *$\mathfrak{M}$ is a free transitive S_3 -set. Thus G_1, G_2, G_3 conventionally designate the three rooted Higgs-response incidences, with no preferred premetric numbering.*

Proof. Precomposition by permutations of $\{1, 2, 3\}$ acts freely and transitively on its bijections to I_3 . The occurrence and response correspondences are canonical and equivariant, carrying the same permutation through all kinds. $\square$

Response meaning of G_1, G_2, G_3 . These designations follow the typed matter fibre and independent rooted response-slot quotient. They name three structural incidences through one carrier, not source tokens, mass order, matrix indices, positive couplings, or three Higgs carriers.

8 Premetric order and residual symmetry

The transported complete generation graph is a triangle, but neither a complete graph nor a rank-one cycle space chooses an ordering of its vertices. This residual symmetry is part of the theorem and remains until additional standing data select a marking.

8.1 Cyclic orientation

Let

$$\text{Mark}_{\text{cyc}}(I_3) := \text{Bij}(\mathbb{Z}/3, I_3).$$

The group $C_3 = \mathbb{Z}/3$ acts by rotating the source marking. Define

$$\text{Cyc}(I_3) := \text{Mark}_{\text{cyc}}(I_3)/C_3.$$

Theorem 8.1 (Cyclic-orientation and start torsors). *The set $\text{Cyc}(I_3)$ has two elements and is a torsor for $S_3/A_3 \cong C_2$. For each cyclic orientation, the possible starts form a C_3 -torsor. Neither torsor has a distinguished element.*

Proof. There are $3! = 6$ bijections from $\mathbb{Z}/3$ to I_3 . The free rotation action has orbits of size three, leaving two cyclic orientations. Even permutations preserve an orientation and odd permutations exchange the two, so the quotient is a free transitive C_2 -set. Once one orientation is fixed, the three rotations give the possible starts and form a free transitive C_3 -set. The full external S_3 action has no invariant orientation or start. $\square$

The coefficient module $\mathbb{Z}$ used in the transported generation cycle quotient admits the nontrivial sign automorphism $\omega \mapsto -\omega$. Rank one chooses a line of circuit residue, not a generator of that line. Orientation reversal therefore remains available until a separate standing witness breaks it.

Corollary 8.2 (No premetric linear hierarchy). *The FCGS-I target supplies no preferred first generation, no preferred cyclic orientation, and no linear mass hierarchy.*

Proof. A linear order on a three-element set chooses a start and an orientation and adds an endpoint relation. Theorem 8.1 leaves the first two choices torsorial, and no premetric relation supplies the third. Numerical mass order lies downstream and cannot be imported to make any choice. $\square$

9 Target-fibred naturality and typed structural codescent

The kinds share a generation and response geometry but retain different full physical fingerprints. Naturality is a same-target theorem; cross-kind agreement factors through an explicitly erased structural reduct.

9.1 Full fingerprints and the maximal lawful category

For an admissible presentation a , write

$$\bar{\kappa}(a) = (r(a), \text{col}(a), q(a), w(a), \chi(a), h(a), H_{\text{br/fix}}, \\ \partial(a), \text{Cont}(a), \text{Src}(a), \Omega^{\text{prim}}(a))$$

for carrier role, color, relative charge, weak component, chirality, kind-specific Higgs incidence, common bridge identity, boundary, continuation, source contracts, and primitive residue coordinate. The last coordinate is the transported joint generation register; it need not equal a species-specific measured CP observable.

Let $\mathcal{K}_f$ be the groupoid of admissible presentations of the fixed kind- f target. Its arrows are all identity-preserving presentation, basis, chart, and continuation transports preserving $\bar{\kappa}$ and carrying complete extensional transports of every source, occurrence, response, registry, status, boundary, continuation, provenance, and failure map consumed by η_f, χ_f . Their identity and composition laws are part of the arrow certificate. Define $\mathcal{K}_{\text{full}} = \coprod_{f \in F} \mathcal{K}_f$.

Theorem 9.1 (Maximal full-target decomposition). *The canonical coproduct inclusion is an equivalence*

$$\coprod_{f \in F} \mathcal{K}_f \xrightarrow{\cong} \mathcal{K}_{\text{full}}, \quad \text{Hom}_{\mathcal{K}_{\text{full}}}(a, b) = \emptyset \quad (a \in \mathcal{K}_f, b \in \mathcal{K}_g, f \neq g).$$

This is maximal among full-target-preserving lawful transports.

Proof. Role exhaustion and pairwise separation put each admissible object in exactly one summand. Neutral and charged colorless roles differ in charge and weak standing; colorless and quark roles differ in color; U, D differ in relative charge, weak component, and $\tilde{H}/H$ incidence. A change is a standing-bearing target update, not a same-target arrow. Conversely the definition admits every identity-preserving presentation, basis, chart, or continuation transport within its fixed fingerprint. No lawful full-target arrow is discarded [11, Secs. 3–4]. $\square$

An absent required transport certificate yields unresolved comparison, not a weaker arrow. Complete extensional transport is already required by the maximal lawful-arrow definition.

9.2 External reindexing and explicit naturality squares

The external group $S_3 = \text{Aut}(K_3)$ acts by $\sigma \cdot (i, r_f) = (\sigma i, r_f)$. Through u_f and rooted-realization disintegration it transports $x \in \mathcal{S}_f(C_f)$, and on response classes $\sigma \cdot \chi_f(x) = \chi_f(\sigma \cdot x)$. These are external remarkings, not physical evolution or internal arrows between different roots.

Let

$$\mathcal{I}, \mathcal{E}, \mathcal{A} : \mathcal{K}_{\text{full}} \longrightarrow \mathbf{Gpd}^{S_3}$$

assign the position object $\text{Disc}(I_3)$, rooted matter fibre $\mathcal{S}_f(C_f)$ (canonically equivalent to the positioned fibre and registered occurrences), and response atlas $\text{Disc}(\mathcal{A}_f)$. A lawful arrow transports their complete signatures.

Theorem 9.2 (Target-fibred naturality on the maximal lawful category). *The root-normalized maps $\eta_f^{I_3} : \text{Disc}(I_3) \rightarrow \mathcal{S}_f(C_f)$ and $\chi_f : \mathcal{S}_f(C_f) \rightarrow \text{Disc}(\mathcal{A}_f)$ assemble as natural isomorphisms*

$$\eta^{I_3} : \mathcal{I} \Longrightarrow \mathcal{E}, \quad \chi : \mathcal{E} \Longrightarrow \mathcal{A}.$$

For every lawful $\alpha : a \rightarrow b$ in $\mathcal{K}_f$,

$$\mathcal{E}(\alpha) \circ \eta_a^{I_3} = \eta_b^{I_3} \circ \mathcal{I}(\alpha), \quad \mathcal{A}(\alpha) \circ \chi_a = \chi_b \circ \mathcal{E}(\alpha).$$

All maps are externally S_3 -equivariant. The positioned-fibre maps $\eta_f : \text{Disc}(\mathcal{E}_f) \rightarrow \mathcal{S}_f(C_f)$ satisfy the corresponding naturality square for the positioned-fibre functor; the two forms are related by the natural inverse pair $\text{Disc}(s_f), \text{Disc}(p_f)$.

Proof. A lawful arrow preserves root, complete occurrence signature, $H_{\text{br}/\text{fix}}$, complete response signature, boundary, continuation, provenance, and failure evaluation. Both paths in the first square give the unique standing occurrence over the transported position; both in the second give the unique zero-truncated standing response class over it. Identity and composition follow from lawful continuation. The same root permutation transported through η, χ gives external equivariance [11, Sec. 4]. $\square$

Each $\mathcal{K}_f$ contains its presentation and continuation transports. The excluded arrows are those between different full physical targets.

9.3 Common-base comparison without full-target identification

The product projections give

$$\theta_{fg}^I = p_g^{-1} p_f : \mathcal{E}_f \longrightarrow \mathcal{E}_g.$$

This preserves generation root and external remarking but forgets the kind-specific fingerprint. It is common-base comparison, not identification of charge, color, chirality, or $\tilde{H}/H$ incidence.

A comparison has exactly one primary status: lawful same-target, presentation-equivalent, common-base comparable, target update, noncomparable at a proved field, or unresolved for a missing certificate. Noncomparability requires proved invariant mismatch. Missing evidence is not a negative theorem, and a target update is not naturality.

9.4 The common premetric structural reduct

Let $D_{\text{gen}}^{\text{open}}$ be the generation-label-free fixed domain on which the root set and direct-comparison graph are formed. **RootSchema** retains rooted incidence and separation; **ResponseSlotSchema** retains structural slot, bridge, and root-preservation data. Let $\partial_0, \text{Cont}_0$ be common boundary evaluation and lawful continuation after kind-specific erasure. The common structural apex is

$$\mathcal{A}_0 = (D_{\text{gen}}^{\text{open}}, I_3, K_3, \Omega^{\text{prim}}, H_{\text{br}/\text{fix}}, \text{RootSchema}, \text{ResponseSlotSchema}, \partial_0, \text{Cont}_0).$$

For each kind, let $\mathcal{A}_f^{\text{str}}$ be its target groupoid retaining exactly these coordinates. Erasure removes kind, charge class, color class, weak representation, chirality, kind-specific Higgs incidence, confinement, terminality, spectra, and metric projection. Let

$$e_f : \mathcal{A}_f^{\text{str}} \longrightarrow \mathcal{A}_0, \quad d_f : \mathcal{K}_f \longrightarrow \mathcal{A}_f^{\text{str}}$$

be literal structural erasure and retained-coordinate restriction. The structural position, occurrence, and slot-class functors use only retained coordinates, so they descend along d_f to

$$\mathcal{I}_f, \mathcal{E}_f, \mathcal{A}_f : \mathcal{A}_f^{\text{str}} \longrightarrow \mathbf{Gpd}^{S_3}.$$

The same coordinate definitions give functors $\mathcal{I}_0, \mathcal{E}_0, \mathcal{A}_0^{\text{rsp}}$ on the apex, with canonical maps $\eta_0^{I_3} : \mathcal{I}_0 \Rightarrow \mathcal{E}_0$ and $\chi_0 : \mathcal{E}_0 \Rightarrow \mathcal{A}_0^{\text{rsp}}$. The superscript on the response functor distinguishes it from the apex itself.

Theorem 9.3 (Common-reduct factorization). *Every e_f is fully faithful and essentially surjective, and*

$$\mathcal{I}_f \cong \mathcal{I}_0 e_f, \quad \mathcal{E}_f \cong \mathcal{E}_0 e_f, \quad \mathcal{A}_f \cong \mathcal{A}_0^{\text{rsp}} e_f.$$

Under these isomorphisms $\eta_f^{I_3}, \chi_f$ are pullbacks of $\eta_0^{I_3}, \chi_0$. The positioned form η_f is obtained through $\text{Disc}(p_f)$ and the same retained-coordinate factorization.

Proof. Every retained coordinate is literally in $\mathcal{A}_0$, and no erased coordinate enters the structural position, rooted fibre, or rooted slot construction. Every local object and arrow has a unique common-core image; every common-core object has a representative in each structural fibre. Equality of common-core arrows is exactly equality on retained coordinates, giving full faithfulness; representatives give essential surjectivity. Factorizations and pullback maps follow coordinatewise [11, Sec. 6]. $\square$

Gravity is not a datum of the apex. Its only role here is the lift-relative guard

$$Y \in \text{Lift}_{\text{adm}}^{\text{AER,P}}(\mathcal{A}_0) \implies \mathbf{G}_{\text{AASC}}(Y).$$

This guard is invariant under erasure and structural transitions. It is not a matter kind, occurrence root, response slot, carrier count, or metric tensor.

10 Effective typed structural codescent

Set

$$X = \coprod_{f \in \mathbf{F}} \mathcal{A}_f^{\text{str}}, \quad p = \coprod_f e_f : X \longrightarrow \mathcal{A}_0, \quad R = X \times_{\mathcal{A}_0} X.$$

The ordered overlap is $\mathcal{A}_{fg} = \mathcal{A}_f^{\text{str}} \times_{\mathcal{A}_0} \mathcal{A}_g^{\text{str}}$. There define $\theta_{fg} = e_g^{-1} e_f$. Literal identity on retained coordinates gives

$$\theta_{ff} = \text{id}, \quad \theta_{gf} = \theta_{fg}^{-1}, \quad \theta_{gh} \theta_{fg} = \theta_{fh}.$$

These transitions preserve η, χ , external S_3 , Ω^{prim} , $H_{\text{br/fix}}$, boundary, continuation, and gravity guard.

Theorem 10.1 (Effective structural-reduct codescent). *The canonical comparison*

$$\text{CoDesc}(R \rightrightarrows X) \longrightarrow \mathcal{A}_0$$

is fully faithful and essentially surjective.

Proof. Every common-core object has a representative in each structural fibre by e_f , giving essential surjectivity. Two local arrows have the same apex image exactly when they agree on the pullback overlap. Inverse and triple-cocycle laws make that agreement independent of the intermediate fibre, so the codescent hom-set is exactly the corresponding apex hom-set. This proves full faithfulness [11, Sec. 8]. $\square$

Only erased structural reducts are identified. Full matter targets stay distinct.

Corollary 10.2 (Tagged structural atlas). *Let $\mathbf{F}_{\text{disc}} = \text{Disc}(\mathbf{F})$ and*

$$\mathcal{A}_{\text{tag}} = \coprod_{f \in \mathbf{F}} \{f\} \times \mathcal{A}_f^{\text{str}} \longrightarrow \mathbf{F}_{\text{disc}}.$$

In the slice over $\mathbf{F}_{\text{disc}}$, $\mathcal{A}_{\text{tag}} \simeq \mathbf{F}_{\text{disc}} \times \mathcal{A}_0$. The kind projection is invariant; no object of one full-target kind fibre is identified with an object of another.

Proof. The e_f identify each deliberately erased reduct with the apex, while retaining tags gives the product equivalence in the slice. Cross-kind overlaps are comparison spans over $\mathcal{A}_0$, not arrows in the tagged atlas. Codescent therefore cannot splice charge, color, chirality, kind-specific Higgs incidence, confinement, terminal, spectral, or metric data [11, Sec. 9]. $\square$

11 Physical realization and conventional particle names

The premetric theorem ends with an unlabelled common base, four typed matter fibres, and their rooted structural response-slot quotients. Conventional particle names belong to a later realization map. Once a marking $m_{\text{phys}} : \{1, 2, 3\} \xrightarrow{\sim} I_3$ has been chosen, one familiar display is

kind fibre	G_1	G_2	G_3	standing type retained
N	ν_e	ν_μ	ν_τ	neutral, colorless, weak matter
L	e	μ	τ	charged, colorless matter
U	u	c	t	up-type colored matter
D	d	s	b	down-type colored matter

The rows have exactly the same three-position generation structure. The entries in the two quark rows are therefore neither exceptional nor generated by a second counting argument. They are the conventional physical names of the three elements of $\mathcal{E}_U$ and $\mathcal{E}_D$. The row signatures remain different: U and D carry distinct relative charge, weak-component, and $\tilde{H}/H$ incidence data, and neither is identified with a colorless row.

The displayed alignment does not choose a premetric first generation. A different marking simultaneously permutes all four rows and their rooted response records. Measured masses may provide an ordering within a separately licensed species-specific numerical projection. Different species need not induce the same order, and no such ordering is a property of I_3 .

11.1 Metric descriptions occur only on admissible lifts

Let $\mathcal{A}_{\text{FCGS}}$ be the completed premetric matter object and $\text{Lift}_{\text{adm}}^{\text{AER,P}}(\mathcal{A}_{\text{FCGS}})$ its admissible physical lifts in the regular-stratum Hamiltonian observable-evaluation comparison class (AER) satisfying presentation arbitrariness (AER-P). This notation asserts no existence theorem; it records the gravity theorem's domain. For such a lift Y , the forgetful map

$$U_{\text{pre}}(Y) = \mathcal{A}_{\text{FCGS}}$$

preserves root base, kind fibres, and zero-truncated rooted slot classes. Metric, causal, field-propagation, spectral, and detector descriptions attach only through further licensed projections. Masses, widths, mixing data, propagation kernels, and detector coordinates leave the original source algebra, Occ_N , I_3 , and primitive fibre cardinalities intact.

Ordinary Standard-Model QFT uses a spacetime background with causal and propagation structure. Here that description, when licensed on an admissible lift, is a metric projection. FCGS-I proves the premetric architecture retained by U_{pre} ; it neither constructs an admissible lift nor replaces QFT dynamics or derives numerical parameters.

12 The predicted heavy neutral lepton is not a fourth generation

The heavy neutral-lepton (HNL) structure enters at a downstream physical-carrier-realization (PCR) target as fixed-base spectral-sector data, not generation-root data. Let $b \in B_8^{\text{ready}}$ and $X \in \text{PhysMod}_b^{E,\text{VJ}}$ be a vectorlike-joint realization satisfying the cited PCR hypotheses. Let $\text{SpecCore}_b(X)$ be its finite set of certified standing-sector objects and $\text{SpecLight}_b(X) \hookrightarrow \text{SpecCore}_b(X)$ its light subset. PCR-6 and PCR-8 give

$$\text{SpecCore}_b(X) \simeq \{S_1(X), S_2(X), S_3(X), S_H(X)\},$$

with three nondegenerate light sectors and one connected heavy sector of internal algebraic rank three [20, Thms. 7.4–7.5, 12.3 and Cor. 16.2] [21, Sec. 2.2 and Thm. P8-23]. In particular,

$$\text{rank}(S_H(X)) = 3, \quad |\pi_0 S_H(X)| = 1.$$

Internal heavy rank is not another root count, and alternate heavy-basis presentations are not extra sectors.

PCR alone does not identify light sectors with the universal generation base. The FCGS-to-PCR interface must contain a licensed fixed-base handoff

$$\lambda_X : I_3 \xrightarrow{\sim} \text{SpecLight}_b(X) = \{S_1(X), S_2(X), S_3(X)\}.$$

This is a downstream marking certificate, neither a new count nor a preferred element of the S_3 torsor. It gives

$$\text{SpecCore}_b(X) = \lambda_X(I_3) \amalg \{S_H(X)\}.$$

Let $\text{Supp}_H^{\text{str}}(X)$ be certified structural heavy-support records and $\text{MetricObs}_b(X)$ licensed metric-observable records. The source-qualified HNL projection factors through typed maps

$$\{S_H(X)\} \hookrightarrow \text{SpecCore}_b(X) \xrightarrow{\rho_H} \text{Supp}_H^{\text{str}}(X) \xrightarrow{\Pi_b} \text{MetricObs}_b(X).$$

Forgetting the spectral extension returns the unchanged base.

Precisely, SpecExt_{I_3} has objects (I_3, Z, λ) with downstream spectral-core set Z and a bijection λ from I_3 to its light subset. Arrows are identity on the first coordinate and typed spectral maps on the second. Set

$$R_X^H = (I_3, \text{SpecCore}_b(X), \lambda_X), \quad U_{\text{root}} : \text{SpecExt}_{I_3} \rightarrow \mathbf{Set}, \quad U_{\text{root}}(I_3, Z, \lambda) = I_3.$$

Theorem 12.1 (Fixed-base HNL nonpromotion). *For b, X, λ_X as above,*

$$\text{SpecCore}_b(X) = \lambda_X(I_3) \amalg \{S_H(X)\}, \quad U_{\text{root}}(R_X^H) = I_3,$$

*and for a putative heavy-root marker $*_H$ there is no injection*

$$\iota_X^{\text{root}} : I_3 \amalg \{*_H\} \hookrightarrow \text{Occ}_N, \quad \iota_X^{\text{root}}|_{I_3} = \text{id}_{I_3}.$$

Thus $S_H(X)$ is one downstream heavy standing sector of internal rank three, not G_4 in the same primitive root inventory.

Proof. Membership in $I_3 = \text{Occ}_N$ uses the primitive architecture and complete neutral occupation receipt of Sections 3–4. The heavy sector is formed only after downstream carrier realization and light-sector handoff. Its invariant is heavy-sector standing with internal rank three, not an additional primitive architectural duty. PCR-6 also proves that spectral visibility or numerical distinction does not promote a sector to a terminal carrier [20, Thm. 12.3]. The extension puts $S_H(X)$ in the spectral coordinate, so U_{root} returns I_3 . Finally $|\text{Occ}_N| = |I_3| = 3$ while $|I_3 \amalg \{*_H\}| = 4$, excluding the displayed injection. Assigning the heavy object additional primitive standing would change its type and the original task, contrary also to unchanged-task nonpromotion [3, Thm. 6.4]. $\square$

A further root of the same primitive occurrence type would have to satisfy the original construction and complete-occupation theorem. The HNL belongs to a different downstream type. Its heavy-sector rank and actual physical content coexist with the primitive three-root inventory.

12.1 Source-certified scouting coordinates

PCR-8 retains the D3 branch benchmark

$$B_{D3} = ([28, 30] \text{ GeV}, c\tau_A^{\text{resp}} = 0.2 \text{ mm}),$$

classified as source-certified and ATLAS-reweight-grid anchored [21, Thm. P8-17 and App. G.1]. D3 names a source target branch, not a dimension or particle label. The 30 GeV point is the direct ATLAS simulated-mass anchor. The 28 GeV endpoint belongs only to the scouting interval; it is no lower pole-mass bound or prediction. The 0.2 mm coordinate is an ATLAS lifetime-reweight-grid point [22], not a proper-lifetime prediction. Neither transfers to the symbolic VJ heavy domain without a certified D3-to-VJ bridge. Physical pole mass and lifetime, full response validation, facility sensitivity, and detector-level discrimination remain separate unresolved questions. Erasing B_{D3} changes no premise or equality in Theorems 4.5, 6.3, or 7.5.

13 Gravity as the necessity condition for physical realization

Gravity is not another particle, interaction boson, or matter kind in the carrier table and does not enter the FCGS count. Within the regular-stratum AER Hamiltonian observable-evaluation class satisfying AER-P, its role is the GR-type canonical constraint architecture required for the same admissible physical-interior target. The imported theorem is confined to that comparison class.

For $Y \in \text{Lift}_{\text{adm}}^{\text{AER,P}}(\mathcal{A}_{\text{FCGS}})$, the gravity theorem gives coisotropic standing, characteristic redescription, local first-class constraint generators, and multiplier-form Hamiltonian presentation [23, Thm. 8.2 and Cor. 8.3]:

$$Y \in \text{Lift}_{\text{adm}}^{\text{AER,P}}(\mathcal{A}_{\text{FCGS}}) \implies G_{\text{AASC}}(Y).$$

A putative lift in that class lacking G_{AASC} fails to instantiate the same admissible physical-interior target.

A metric-background QFT is downstream: its spacetime, causal, and propagation standing are evaluated on an admissible physical lift with that constraint architecture. Gravity is a necessary property of every such lift. Necessity proves neither existence of a lift nor sufficiency of gravity.

Proposition 13.1 (Gravity no-backflow). *The gravity necessity interface creates or identifies no element of I_3 , changes no kind signature, and supplies no Higgs-response incidence. It therefore changes neither exact three nor the four-fibre conclusion.*

Proof. All defining root data lie at the primitive architecture and occupation stage. Gravity is evaluated on a later admissible lift Y with $U_{\text{pre}}(Y) = \mathcal{A}_{\text{FCGS}}$. Its conclusion is a necessary property of Y , with no map into Occ_N , relation on I_3 , change of $\mathfrak{F}$, or new bridge carrier. The forgetful map and strict dependency order in Section 2 forbid backflow. $\square$

This imports no Einstein equations, unique metric, gravitational coupling, ADM or hypersurface-deformation dynamics, refoliation invariance, QFT propagation law, or realization functor. Within the regular-stratum AER-P class, a lift lacking the gravity role fails the same physical-interior target.

14 Counterexample criteria and dependency protections

The theorem has explicit failure conditions when a standing-bearing type is erased or dependency is reversed.

Table 2: Structural falsifiers for the FCGS-I endpoint.

Prohibited alteration	Conclusion invalidated
Define Occ_N from a component count	Independent occurrence formation and complete occupation receipt.
Insert three family names, a fixed 3×3 dimension, or measured mass before the count	Count-independent primitive derivation.
Use rank one as a premise of the primitive count	Primitive-standing derivation and consequent rank.
Identify U with D	Relative-charge, weak-component, and $H/\tilde{H}$ separation.
Use the four-kind bundle to prove the base cardinality	Dependency order and generation-cardinality receipt.
Let response equivalence or an internal arrow change occurrence root	Exact rooted response correspondence.
Promote operator availability to positive numerical coupling	Typed status and positivity distinction.
Multiply $H_{\text{br}/\text{fix}}$ with its incidence legs	One-carrier identity.
Call θ_{fg} a full-target equivalence	Target-fibred naturality and kind separation.
Import erased charge, color, chirality, terminal, spectral, or metric fields into structural codescent	Common-reduct codescent and tagged preservation.
Promote S_H to an extra unchanged-task primitive root	Fixed-base HNL nonpromotion.
Let metric or gravity data alter Occ_N	Premetric priority and no-backflow.

Primitive cardinality is carried by joint adequacy, necessity, and complete compatible occupation. Quark/lepton common geometry follows by exact base change. Response cardinality comes from the independently formed rooted slot quotient; HNL and gravity conclusions retain their separate typed hypotheses.

Theorem 14.1 (FCGS-I endpoint). *For the fixed premetric Standard-Model matter target,*

$$I_3 = \text{Occ}_N, \quad |I_3| = 3, \quad \mathcal{E}_{\text{SM}} = I_3 \times \{r_N, r_L, r_U, r_D\},$$

with pairwise distinct role types. On complete receiving interfaces satisfying F1–F9, H1–H7, and R1–R9, there are canonical S_3 -equivariant equivalences

$$\text{Disc}(I_3) \xrightarrow{\eta_f^{I_3}} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} \text{Disc}(\mathcal{A}_f) \quad (f \in \mathbf{F}).$$

All kinds share one unlabelled three-root generation geometry and exactly three rooted structural Higgs-response-slot incidences each, through one bridge/fixation carrier. Only the positive-witness subquotient may be smaller. The G_1, G_2, G_3 markings are conventional, with no preferred premetric order. Position, matter-fibre, and response constructions are natural on the maximal lawful full-target category. Explicitly erased structural reducts have effective codescent to the common apex while kind tags remain distinct.

Proof. Theorem 4.5 supplies the primitive base and neutral receipt. Proposition 5.1 gives four count-free distinct roles; Theorem 6.3 forms their exact post-count fibres. Theorems 6.5, 6.6, and 6.7 give source-certified occurrence fibres and standing correspondence under their F/H contracts. Theorem 7.5 supplies response correspondence under R1–R9; Theorem 7.4 assembles the rootwise structural witnesses

and licensed connectivity supplied by R4 and R6, retaining the common carrier. Theorems 8.1, 9.2, 10.1, and Corollary 10.2 supply symmetry, transport, structural codescent, and tagged preservation. Each conclusion follows from its typed antecedent. $\square$

This is the direct endpoint. Theorem 12.1 separately concerns PCR realizations satisfying readiness and fixed-base handoff. Proposition 13.1 is the inherited necessity for admissible regular-stratum AER-P lifts. Neither enlarges the endpoint or supplies one of its premises.

15 Conclusion

The original generation duties determine a minimal jointly adequate primitive architecture. Complete compatible occupation and the source-qualified neutral receipt give the common three-root base I_3 . Its rank-one generation cycle register follows from that construction.

The generation-erased Standard-Model profile supplies four role types. Generation copies of that profile form $I_3 \times \{N, L, U, D\}$, with common generation geometry and distinct physical signatures. On the complete positioned and rooted-realization receiving interfaces, the transfer and disintegration theorems certify the actual occurrence fibres.

With the source witness and connectivity contracts, the response module forms and quotients complete arenas over those occurrences, proving three structural slots per kind through one bridge/fixation carrier. Positive witnesses remain separately typed. Physical names, masses, mixing, search benchmarks, and detector response follow only through licensed projections. Gravity is necessary on admissible regular-stratum AER-P lifts, on which metric QFT may have standing.

The resulting architecture preserves its original primitive work under downstream enrichment without erasing additional physical content. It distinguishes primitive roots, typed role positions, response slots, positive witnesses, and terminal or spectral objects throughout.

A Theorem-interface ledger

Every import retains its stated target and hypotheses. Complete source contracts are inputs to the receiving theorems, not consequences of the desired component count.

Interface	Source location	Imported conclusion
Live residue	Role-Occupancy Closure, Defs. 3.9–3.18; Thms. 7.2–7.4	Eligibility, jointly compatible preassignments, live residue, and occupation are formed at one fixed certificate and target. Cardinality equality is not compatibility.
Neutral use algebra	Present Theorem 3.1; Standing Carrier Identity Closure	Licensed uses factor functorially through the complete groupoid-valued signature; natural isomorphism under all uses agrees with natural isomorphism of complete signatures.
Core and congruence	Present Theorems 3.2–3.3	The finite incidence-minimal active core is canonical relative to its presentation code. Retained nonanchors are deletion-essential; all uses, continuations, boundaries, and failures descend before standing-component inspection.
Presentation identity	Standing Carrier Identity Closure, Defs. 3.6–3.7; Thms. 3.9, 4.2, 4.4	Complete target-relative continuation signatures induce the same-scope identity quotient at the frozen interface. Distinct rooted incidences are not thereby identified.
Primitive cardinality	Generation Cardinality, Def. 6.1; Thms. 5.4, 6.2, 6.4, 6.6; Cor. 5.5; present Sections 3–4	Joint primitive adequacy with no proper adequate support gives exactly three architectural positions. Complete compatible occupation gives actual root count. Rank one follows. Unchanged-task enrichment adds no primitive standing; the source-defined neutral occupation is separately proved.
Matter roles	Matter Carriers, Defs. 5.1, 5.4–5.6, 5.9; Thms. 14.2, 19.3–19.11	Representation-response, color-bearing, charged-colorless, neutral-weak, and mass-response-slot roles have primitive standing. This classifies roles, not their copy count.
Charge separation	Charge Lattice, Lemmas 9.1–9.2; Thm. 9.3	Relative one-family hypercharges are unique up to common normalization and global sign, separating U, D signatures.
Generation semantics	Generation Count and Residual Parameters, Def. 9.1; Prop. 9.2	Cardinality counts copies of a typed representation-family profile. Changing it changes role and coupling-slot inventories, not presentation skin. The recorded empirical count is erased.

Interface	Source location	Imported conclusion
Structural tuple	Standalone Standard Model Structure, Defs. 1.7, 1.9, 1.12; Thms. 1.13, 9.7	Erasing N_g , family labels, matrix dimensions, and indexed numerical coordinates leaves the chiral profile, bridge identity, and typed operator ledger. Neither source count nor per-root coverage is imported.
Positioned transfer	Family-Cell Transfer and Exact3, hypotheses F1–F9 and proof	F4 requires one compatible live occupant per positioned cell and F5 requires complete occurrence coverage. Under these and the other source contracts, the theorem proves an equivariant occurrence bijection. It performs no second kindwise count and does not discharge F4/F5.
Rooted realization	Rooted-Realization Comparison, hypotheses H1–H7 and proof	H4 supplies realization coverage and H5 supplies rooted-transport completeness. Under them the raw fibres are nonempty and connected and zero truncation equals the occurrence set. Raw isotropy is not removed by a claim about components.
Carrier exhaustion	Standard Model Carrier Exhaustion, Thms. 17.1, 20.1	The fixed package is complete at its stated primitive-role scope; this is not a rooted-occurrence count.
Higgs identity	Higgs Bridge, Defs. 4.1–4.3, 7.1, 8.1; Thms. 7.4, 7.6, 8.4, 9.2, 12.1; Higgs Identity, Thms. 6.6, 7.3, 8.1, 14.4; Cor. 8.2	The bridge/fixation role has one carrier identity. Multiple response legs do not create extra Higgs carriers.
Response quotient	Independent Rooted Response-Slot Quotient, statement, hypotheses, proof; Rooted Higgs-Response Source-Use Contracts; Yukawa Compatibility, Defs. 3.3–3.4, 5.1, 5.3, 6.1–6.2; Thms. 5.4–5.5, 6.3–6.5, 7.1, 26.1	Under R1–R9, complete raw arenas precede reduction. R4 supplies an admitted structural slot over each root; R6 supplies connectivity of each entire licensed standing root fibre. The theorem derives the component bijection from those inputs. Neither carrier/classifier uniqueness nor the citation itself discharges them. Positivity needs its separate witnessed branch.
Naturality	Target-Fibred Naturality and Typed Common-Base Codescent, Secs. 3–6	Position, standing-occurrence, and rooted-response maps are natural isomorphisms on the maximal lawful full-target category. Cross-kind agreement uses explicit structural erasure.
Codescent	Same source, Secs. 8–9	Groupoid codescent of erased reducts gives the common premetric apex. The tagged atlas keeps four kinds and distinct full targets.

Interface	Source location	Imported conclusion
Heavy sector	Physical Nature and Minimal Identity, Thms. 7.4–7.5, 12.3; Cor. 16.2; Neutrino Carrier Prediction, Thms. P8-17, P8-23	A VJ PCR realization satisfying readiness has three light standing sectors and one connected heavy sector of internal rank three. Identification with I_3 requires fixed-base handoff. D3 28–30 GeV and 0.2 mm are a scouting benchmark, not a VJ pole-mass or lifetime derivation.
Gravity	Gravity as Structural Necessity, Thm. 8.2; Cor. 8.3	Every admissible regular-stratum AER physical lift satisfying AER-P instantiates GR-type canonical constraint architecture. Existence and sufficiency are not inferred.

Hypothesis correspondence. The positioned-transfer source’s F1–F9 are: prior base derivation; count-free role-cell formation; complete ledgers; unique positioned occupancy; exhaustive coverage; weak-pair coherence; neutral-anchor agreement; internal root separation with external S_3 transport; simultaneous source-cell compatibility. They correspond in order to Theorem 6.5.

The rooted-comparison source’s H1–H7 are: prior activity and congruence; fixed-target restriction; root transport; registry completeness; rooted-transport completeness; root separation; external symmetry. They correspond in order to Theorem 6.6. The response source’s R1–R9 are: prior root result; one bridge identity; complete preselection registries; admitted structural slot existence; canonical factorization; rooted congruence and connectivity; root separation; total ordered branch classification; projection separation. In particular, R4 and R6 are receiving obligations used by Theorems 7.4–7.5. These labels identify source hypotheses, not extra object coordinates. The certificates prove the conditional transfer and component statements; they do not independently discharge F4/F5, H4/H5, or R4/R6.

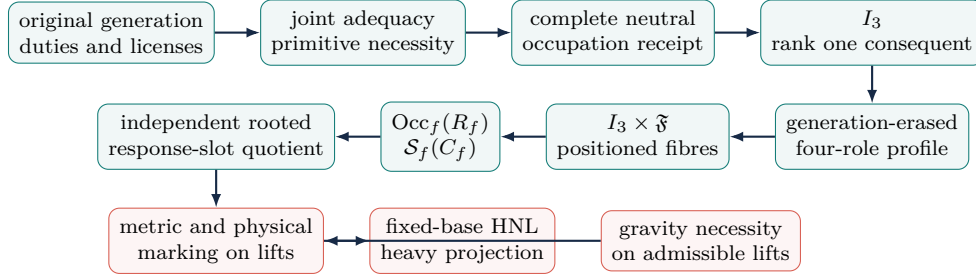
Three numerical uses remain distinct: $|I_3| = 3$ is primitive root cardinality; $|\mathcal{E}_f| = 3$ is typed base change; $\text{rank}(S_H(X)) = 3$ is internal heavy rank. Only the first supplies the primitive generation count.

B Notation and object types

Symbol	Meaning
T_N^-	Frozen premetric neutral receiving target before family and spectral data.
C_N	Complete certificate: target, semantic uses, activated signature, compatibility and failures.
$\mathcal{P}_N$	Count-blind active neutral presentation groupoid.
Occ_N	Rooted neutral incidence inventory formed before components and received by complete occupation of primitive generation work.
$\mathcal{R}_N^{\text{vert}}(C_N)$	Raw vertical neutral realization groupoid over $\text{Disc}(\text{Occ}_N)$.
$\mathcal{S}_N(C_N)$	Structural zero-truncation of the raw vertical groupoid, equivalent to $\text{Disc}(\text{Occ}_N)$.
I_3	Derived unlabelled root base after $ \text{Occ}_N = 3$ is proved.
$\mathfrak{F}$	Generation-erased role profile $\{r_N, r_L, r_U, r_D\}$.
$\mathcal{E}_f, \mathcal{E}_{\text{SM}}$	Typed position fibre $I_3 \times \{r_f\}$ and full matter bundle $I_3 \times \mathfrak{F}$.
R_f, C_f	Positioned realization target and complete activated interface: uses, signatures, compatibility, continuation, boundary, failures.
$\text{Occ}_f(R_f)$	Post-count source-certified occurrences from live positioned cells.
$\mathcal{R}_f(R_f)$	Rooted-realization groupoid of complete presentations with registered incidence.
$\mathcal{S}_f(C_f)$	Zero truncation, equivalent to $\text{Disc}(\text{Occ}_f(R_f))$.
p_f, s_f	Projection $\mathcal{E}_f \rightarrow I_3$ and canonical inverse insertion.
u_f	Exact positioned-occupant bijection $\mathcal{E}_f \rightarrow \text{Occ}_f(R_f)$.
$\eta_f, \eta_f^{I_3}$	Canonical $\text{Disc}(\mathcal{E}_f) \rightarrow \mathcal{S}_f(C_f)$ equivalence and root-normalized composite with $\text{Disc}(s_f)$.
$H_{\text{br}/\text{fix}}$	Unique bridge/fixation carrier.
$Y_f(o)$	Complete raw response arena formed before class selection.
$H_f(o)$	Nonempty connected standing response fibre under complete rooted response equivalence.
Symbol	Meaning
Q_f^{slot}	Coproduct of standing root fibres, with three components per kind.
$Q_{f,\text{acc}}^Y$	Source zero-truncated quotient, equivalent to $\text{Disc}(\pi_0 Q_f^{\text{slot}})$.
$\mathcal{A}_f$	Zero-truncated rooted response atlas $\pi_0 Q_f^{\text{slot}}$.
χ_f	Exact response equivalence $\mathcal{S}_f(C_f) \rightarrow \text{Disc}(\mathcal{A}_f)$.
Q_f^+	Positive-witness subquotient, at most three components with separate positivity obligations.
$\mathcal{A}_0$	Common apex retaining $D_{\text{gen}}^{\text{open}}, I_3, K_3, \Omega^{\text{prim}}, H_{\text{br}/\text{fix}}$, root/slot schemas, $\partial_0, \text{Cont}_0$.
e_f	Fully faithful essentially surjective erasure from the kind's structural reduct to the apex.
S_H	One downstream fixed-base heavy sector of internal rank three.
SpecExt_{I_3}	Fixed-base spectral extensions; U_{root} returns I_3 .
GAASC	Necessary gravity role on admissible regular-stratum AER-P physical lifts.

C Proof-dependency graph

The count and receiving chain have no backward arrows:



Generation erasure precedes profile instantiation. Compatible occupancy and rooted-realization disintegration establish the occurrence fibres before response quotienting. Gravity constrains admissible lifts without an arrow into primitive count or bundle.

D Structural invariance and counterexample criteria

Duplicate presentation and root permutation preserve the construction. A changed signature coordinate used by a licensed continuation changes the complete-signature class; pure presentation skin does not.

The four-kind theorem needs no family-name table. Proved I_3 and role signatures form the fibres uniformly. Role deletion, signature duplication, using the bundle to prove the base count, or changed relative U/D charges cause type or dependency failure.

Complete raw arenas and failure ledgers are total. Only complete licensed objects enter the standing quotient, with one nonempty connected fibre over each registered occurrence. The positive subquotient may be empty. Interface-incomplete proposals remain failures and are not promoted. Root erasure destroys the component correspondence, so is not an admissible simplification of Theorem 7.5. These criteria supplement the proofs.

Source status. The cited companion manuscripts supply the interfaces above with their explicit domains and license obligations. The count uses Theorem 6.2 of the primitive-standing reference edition, together with the local complete-occupation and exact-three results.

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